

MULTI-CLASS DISEASE DIAGNOSIS USING MULTINOMIAL LOGISTIC REGRESSION

Abstract

Early and reliable disease diagnosis is an important task in healthcare because symptoms and patient characteristics can vary considerably between individuals. Traditional diagnosis generally depends on clinical expertise and the interpretation of several patient attributes. Machine learning can assist this process by learning patterns from previously observed patient records and producing a predicted disease class. This study presents a multi-class disease diagnosis approach using Multinomial Logistic Regression. The proposed system uses the Disease Symptoms and Patient Profile Dataset and considers symptoms and patient information including Fever, Cough, Fatigue, Difficulty Breathing, Age, Gender, Blood Pressure and Cholesterol Level. Categorical variables are encoded, numerical features are prepared and scaled where required, and the data is divided into training and testing subsets. Multinomial Logistic Regression is then trained to estimate the probability of multiple disease classes and assign the most probable class to a patient record. The work follows a machine-learning pipeline consisting of data preparation, exploratory data analysis, feature preparation, model training, prediction and evaluation. The approach provides a simple and interpretable baseline for multi-class medical prediction and can be extended using larger clinical datasets, additional symptoms, laboratory measurements and advanced classification algorithms.

Keywords: Machine Learning, Multi-Class Classification, Disease Diagnosis, Multinomial Logistic Regression, Healthcare, Medical Prediction, Patient Symptoms

1. Introduction

Disease diagnosis involves identifying a medical condition from information collected about a patient. Symptoms, demographic characteristics and clinical measurements provide important evidence for distinguishing one disease from another. As healthcare datasets become increasingly available, machine learning methods can be used to discover relationships within patient records and support diagnostic decision-making. Such systems are intended as decision-support tools and do not replace qualified medical professionals.

Many classification problems in healthcare are binary, where a model determines whether a condition is present or absent. However, real diagnostic settings may involve several possible diseases. This makes multi-class classification important because a patient may need to be assigned to one disease category from a set of possible classes.

This research investigates Multinomial Logistic Regression for multi-class disease diagnosis. Logistic regression is widely used for classification because it provides probabilistic predictions and relatively interpretable model parameters. The multinomial form extends the method to situations in which the target contains more than two classes.

1.1 Objectives

- To prepare a patient dataset containing symptoms and demographic/clinical attributes for machine-learning classification.
- To perform exploratory analysis of disease classes and input features.
- To encode categorical variables and prepare numerical variables for modelling.
- To develop a Multinomial Logistic Regression model for predicting multiple disease categories.
- To evaluate the model using standard classification measures such as accuracy, precision, recall and F1-score.

2. Related Work

Machine learning has been investigated extensively for medical diagnosis and clinical prediction. Logistic regression is one of the commonly used statistical and machine-learning approaches because its coefficients can be interpreted in relation to the estimated probability of an outcome. For multi-class outcomes, multinomial logistic regression can estimate probabilities for several mutually exclusive classes.

Previous studies have applied multinomial logistic regression to medical classification problems involving multiple diagnostic categories. Research on thyroid disorders, for example, has compared multinomial logistic regression with neural-network approaches for classifying disease status. Other work has investigated patient-facing diagnostic models in which symptoms are used to estimate multiple possible diagnoses.

The present work uses the Disease Symptoms and Patient Profile Dataset with symptom, demographic and clinical-profile features. The focus is on constructing a clear end-to-end machine-learning pipeline using Multinomial Logistic Regression.

3. Background

3.1 Machine Learning for Disease Diagnosis

Supervised machine learning learns a mapping between input features and known target labels. In disease diagnosis, the input may contain symptoms, age, gender and clinical measurements, while the target represents the disease category.

3.2 Multi-Class Classification

Multi-class classification refers to a problem in which an observation belongs to one of more than two possible classes. If the target contains K disease categories, the classifier estimates the probability associated with each category and selects the class with the highest predicted probability.

3.3 Multinomial Logistic Regression

Multinomial Logistic Regression extends logistic regression to K classes. For an input vector x , the model estimates a probability for each class using a softmax function.

$$P(Y = k | x) = \exp(\beta_k^T x) / \sum_{j=1}^K \exp(\beta_j^T x)$$

3.4 Data Pre-processing

Pre-processing converts raw patient records into a representation suitable for machine learning. Typical operations include checking missing values, handling inconsistent entries, encoding categorical variables and scaling numerical variables.

4. Analysis

4.1 Dataset Description

The project uses the **Disease Symptoms and Patient Profile Dataset**. The selected predictors are Fever, Cough, Fatigue, Difficulty Breathing, Age, Gender, Blood Pressure and Cholesterol Level. The target variable is Disease.

4.2 Exploratory Data Analysis

EDA is performed to understand data types, missing values, duplicate records, descriptive statistics and disease-class distribution. Feature-level analysis can also be used to observe how symptoms and patient characteristics vary across the dataset.

4.3 Performance Evaluation Measures

The classifier can be evaluated using accuracy, precision, recall and F1-score. Macro and weighted averages provide additional information for datasets containing multiple disease classes. A confusion matrix shows which disease categories are confused with one another.

5. Materials and Methods

5.1 Proposed Workflow

Dataset collection → data inspection → exploratory data analysis → preprocessing → categorical encoding → feature scaling where appropriate → train-test split → multinomial logistic regression training → prediction → evaluation.

5.2 Data Preparation

The Disease column is separated as the target variable, while the selected symptoms and patient-profile attributes form the input matrix. Categorical variables are transformed into numerical representations.

5.3 Feature Scaling

Feature scaling is useful when numerical variables operate on different ranges. Standardization can be applied so that numerical features are placed on a comparable scale before model fitting.

5.4 Train-Test Split

The prepared dataset is divided into training and testing subsets. The training subset is used to estimate model parameters, while the testing subset is reserved for evaluation.

5.5 Model Training

Multinomial Logistic Regression is trained using the prepared feature matrix and disease labels. The model returns

class probabilities and the disease class with the maximum probability is selected.

5.6 Implementation Environment

The implementation can be carried out in Python using pandas, NumPy, Matplotlib/Seaborn and scikit-learn.

6. Results and Discussion

The model evaluation should be reported using the actual test-set results obtained from the project implementation. The main outputs are the confusion matrix and classification report containing precision, recall and F1-score for each disease class, together with overall accuracy.

Because the dataset contains a large number of disease categories and some categories have relatively few examples, accuracy alone may not fully describe model behaviour. Per-class recall and F1-score are particularly useful for identifying classes that the model predicts less reliably.

The multinomial logistic model provides an interpretable baseline. However, prediction quality depends strongly on the representativeness, size and reliability of the training data. The results should therefore be interpreted as machine-learning predictions rather than clinical diagnoses.

7. Challenges and Future Research

Several challenges can affect multi-class disease diagnosis. A dataset with many disease classes may contain limited observations for individual classes, creating class imbalance and reducing the ability of the model to learn class-specific patterns. Symptoms can also overlap across diseases.

Future improvements may include:

- Increasing the size and diversity of the patient dataset.
- Adding laboratory measurements, medical history and additional clinically relevant symptoms.
- Applying class-balancing or resampling strategies when appropriate.
- Comparing multinomial logistic regression with Random Forest, Support Vector Machine, gradient-boosting and neural-network models.
- Performing systematic hyperparameter tuning and cross-validation.

8. Conclusion

This study presents a machine-learning approach for multi-class disease diagnosis using Multinomial Logistic Regression. The proposed pipeline uses patient symptoms and profile information, including Fever, Cough, Fatigue, Difficulty Breathing, Age, Gender, Blood Pressure and Cholesterol Level, to predict a disease category.

Multinomial Logistic Regression is suitable as an interpretable baseline for a problem in which the target contains multiple mutually exclusive disease classes. Future research can strengthen the system by using larger and more representative clinical datasets, adding richer patient information and comparing multiple machine-learning approaches.

Declarations

Author contribution statement: The project work includes dataset preparation, exploratory analysis, machine-learning implementation and manuscript preparation.

Funding statement: No specific external funding was used for this academic project.

Competing interest statement: No competing interests are declared.

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